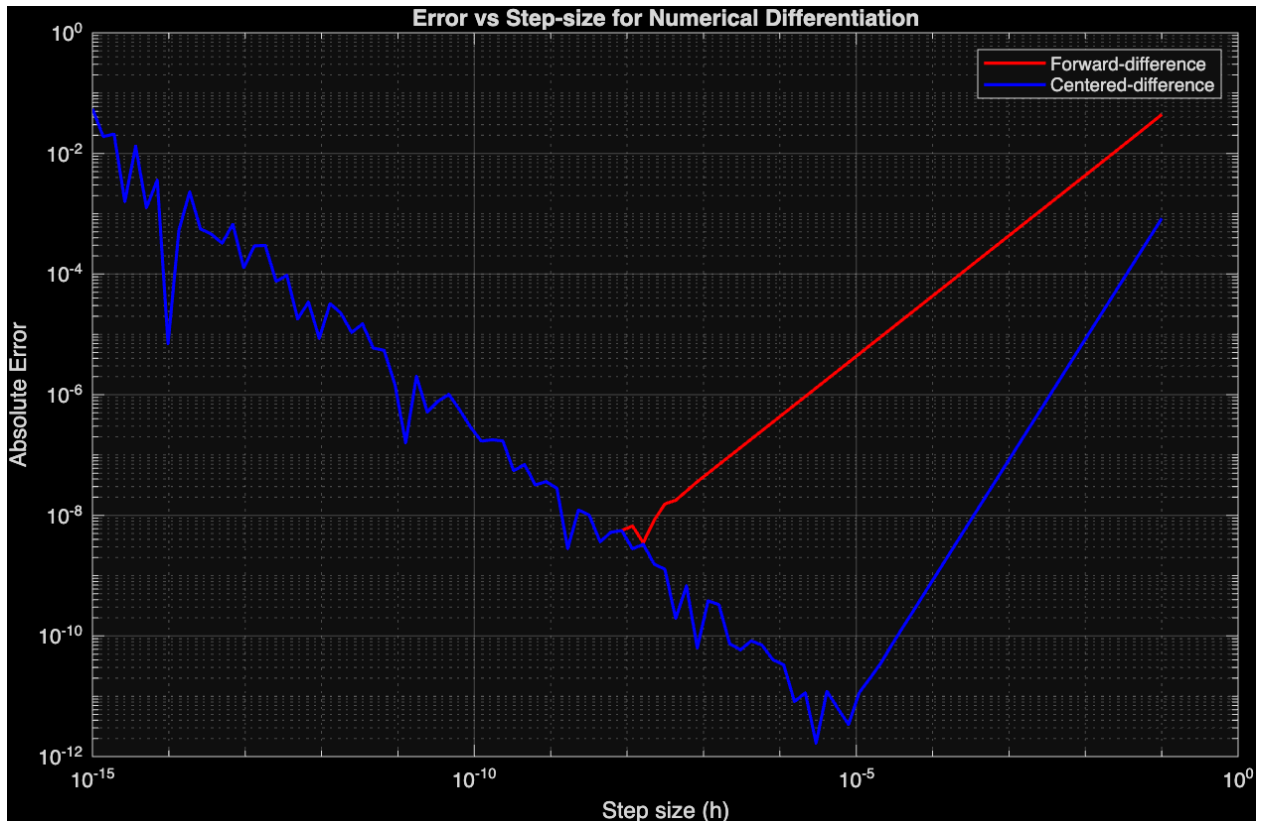




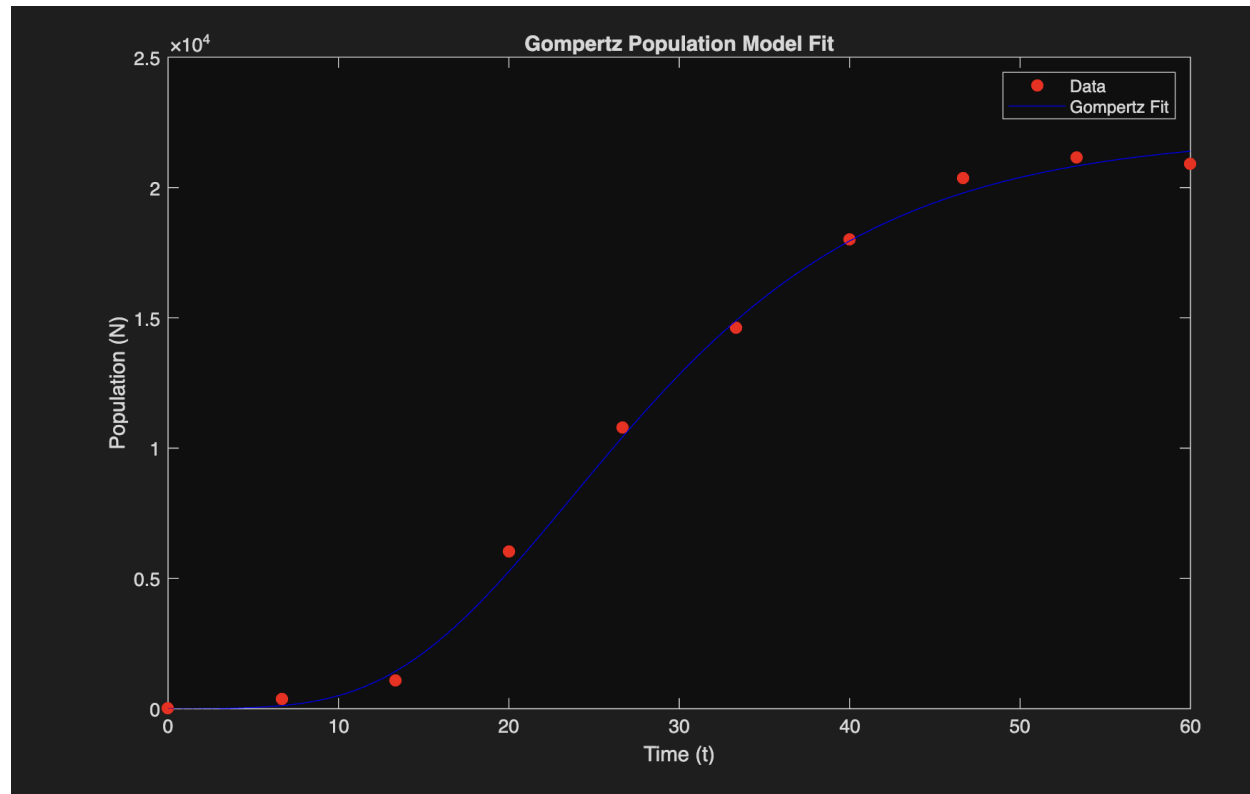
2)

- a) Forward-difference: The optimum h is $10e-8$. At this point, the error is $10e-8$.
- b) Centered-difference: The optimum h is $10e-5$ to $10e-6$. The absolute error is significantly lower, reaching $10e-12$.
- c) Comparison: The centered-difference formula is more accurate because its truncation error is $O(h^2)$, whereas the forward-difference is $O(h)$. However, both fail at very small h due to floating-point round-off error

3) 1

$$\begin{aligned}
 & \text{3a-} \ln(N(t)) = \ln(1 - \exp(-bt)) \\
 & \frac{\ln(N(t))}{\ln(1 - \exp(-bt))} = 1 \\
 & 1 - \frac{\ln(N(t))}{\ln(1 - \exp(-bt))} = \exp(-bt) \\
 & \ln\left(1 - \frac{\ln(N(t))}{\ln(1 - \exp(-bt))}\right) = -bt \\
 & y = \ln\left(1 - \frac{\ln(N(t))}{\ln(1 - \exp(-bt))}\right)
 \end{aligned}$$

a)



b) Estimated b: 0.097222

c) 10^{-5} because 10^{-1} is too large and leads to high truncation error, while 10^{-9} is too small for this formula and is susceptible to round off error.

h	Numerical Deriv	Error
$1.000000e-09$	674.543116	$1.493678e-02$
$1.000000e-05$	674.536157	$7.977599e-03$
$1.000000e-01$	674.532644	$4.464639e-03$

d)